Exploring Macro-Level Factors Associated with High School Non-Completion in Ontario: A PESTEL-Based Analysis

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Executive Summary

This report presents an exploratory analysis of how macro-level political, economic, and social conditions relate to high school non-completion in Ontario between 2018 and 2022. Using a PESTEL-informed approach, the analysis focuses on identifying consistent patterns, directional relationships, and practical insights to support decision-making, rather than establishing causality. Given the small dataset of five annual observations, the findings are interpreted cautiously, but they consistently point to one central idea: high school non-completion is not an isolated educational issue, being deeply embedded in the broader external environment.

Across descriptive statistics, trend analysis, correlations, regression models, and comparative analysis, a clear pattern emerges. Dropout rates are not driven by a single factor, but by the interaction of multiple external conditions. Among these, the economic dimension shows the strongest and most consistent relationships, followed by political factors, while social variables provide important context but weaker explanatory power.

Key Findings

Economic conditions stand out as the most influential dimension. Higher median income is consistently associated with lower dropout rates, while indicators of economic stress, such as poverty and youth unemployment, align with worse outcomes. This pattern holds across all stages of the analysis, and the economic model provides the strongest standalone fit. When considered together, these results suggest that household financial stability is deeply connected to student retention, shifting the framing of dropout from a purely educational issue to one rooted in broader socio-economic conditions.

Political factors also show meaningful influence. The period following the identified policy reform (2021–2022) coincides with lower dropout rates, alongside improvements in funding per student and attendance. While this does not establish causality, the evidence suggests that policy

stability and sustained investment create more favourable conditions for student success. Notably, structural policy shifts appear to carry more explanatory weight than short-term fluctuations in funding alone.

Within the social dimension, attendance emerges as a particularly important and actionable factor. It is consistently associated with lower dropout rates and functions both as an early warning signal and a direct point of intervention. At the same time, broader indicators (such as the percentage of adults without a high school diploma) highlight longer-term, intergenerational influences that shape student outcomes beyond the immediate school environment.

When considered together, the combined regression model provides the strongest overall fit, reinforcing the idea that dropout rates are best understood as the result of overlapping forces rather than isolated drivers. Comparative analysis further supports this interpretation, showing that periods characterized by stronger economic conditions and more supportive policy environments consistently align with better student outcomes.

Strategic Takeaways

From a decision-making perspective, the findings point to several important implications. Most fundamentally, education policy cannot be effective if treated in isolation. The strong link between economic conditions and student outcomes means that efforts to reduce dropout must extend beyond school-level interventions and be coordinated with broader economic and social policies. Education systems need to be aligned with wider policy frameworks that address household stability and community conditions.

At the same time, the analysis highlights the importance of policy consistency and long-term investment. Stable and adequately funded systems appear better positioned to sustain improvements, while frequent shifts in direction or reductions in funding risk undermining progress, particularly during periods of economic stress.

Attendance stands out as one of the most immediate and actionable levers available. Because it is both a strong predictor of dropout and a factor that schools can monitor in real time, targeted strategies to improve attendance offer a practical way to identify and address early signs of disengagement.

More broadly, the findings suggest that risks and opportunities in the education sector tend to reinforce each other. Economic hardship, policy instability, and adverse social conditions can combine to increase dropout risk, while improvements across these dimensions can create a cumulative positive effect. This points to the need for integrated, system-level strategies that move beyond fragmented or school-centric approaches.

Methodology and Limitations

The analysis is based on a small, curated dataset drawn from Statistics Canada, covering Ontario from 2018 to 2022. The methodology follows a stepwise approach, combining descriptive statistics, trend analysis, correlations, scatter plots, and regression models, structured through the PESTEL framework to support interpretation.

These findings should be interpreted with caution due to several important limitations. The small sample size (n=5) limits statistical power and makes the results sensitive to individual years, particularly atypical ones such as 2020. The use of aggregated provincial data may mask regional variation and introduce aggregation bias. In addition, the presence of multicollinearity and potential endogeneity makes it difficult to isolate individual effects.

The analysis is also constrained by data availability, as technological, environmental, and legal dimensions of the PESTEL framework are not included. As a result, the findings should be understood as exploratory and directional rather than definitive evidence.

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1. Introduction

Education is widely recognized as a cornerstone of individual opportunity and societal prosperity. In Ontario, the richest province of Canada, while most students successfully complete high school, a persistent portion of the population leaves the education system without a diploma, facing long-term consequences in terms of employment, earnings, and overall well-being. As a result, high school non-completion remains an ongoing concern nation-wide, as it impacts not only on individual life trajectories, but also brings broader social and economic implications.

The issue is often approached from a school-centric perspective, with emphasis placed on curriculum design, teaching quality, and student-level support. While these elements are undeniably important, they do not fully capture the environment in which educational outcomes are shaped. Schools and students do not operate in isolation; their performance and engagement are influenced by a wider set of external conditions, including the policy landscape, the financial stability of households, and the broader social context in which they are embedded. These macro-level forces can either create supportive conditions for success or increase the risk of disengagement and dropout.

This report presents an exploratory analysis of how such external factors relate to high school non-completion in Ontario between 2018 and 2022. Adopting a PESTEL-informed framework (focusing on Political, Economic and Social components), the study moves beyond isolated explanations to examine how multiple conditions evolve together and align with changes in dropout rates over time. Drawing on a curated set of indicators from Statistics Canada (namely median income, policy reforms and funding, attendance patterns, and adult educational attainment), the analysis aims at reframing the dropout issue within a broader, interconnected system.

It is important noticing that, given the structure and size of the dataset, the objective is not to establish causality, but rather to identify meaningful patterns, directional relationships, and consistent signals that can inform decision-making. In order to support this, the report combines descriptive analysis, trend exploration, regression techniques, and comparative analysis. These methods are used in a complementary way, allowing findings to be interpreted with greater confidence despite data limitations.

The results are ultimately translated into an environmental impact perspective and a set of strategic insights, with the goal of supporting more informed, system-level responses within the education sector, by highlighting key risks, uncovering potential opportunities, and contributing to a more nuanced understanding of the conditions that shape student outcomes.

2. Methodology

This study was designed as an exploratory analysis of how broader external conditions may relate to changes in high school non-completion in Ontario. Rather than aiming to produce a large-scale predictive model or make strong causal claims, the purpose was to build a clear and interpretable picture of how selected indicators moved over time and how they may have aligned with shifts in student outcomes. Given the short time span and the structure of the dataset, the methodological approach prioritized consistency, readability, and cautious interpretation throughout the analysis.

2.1 Data Sources and Scope

This analysis uses a small, structured dataset covering Ontario from 2018 to 2022. The data were drawn primarily from publicly available Statistics Canada sources and consolidated into a single analytical file. From the beginning, the project was conceived as a big-picture study of how selected political, economic, and social conditions may be associated with changes in high school non-completion over time. The original proposal defined the main outcome variable, identified the key Statistics Canada sources, and established the initial PESTEL-based framing for the study.

The final dataset is organized at the yearly level, with one observation per year. This naturally limits the level of detail that can be captured, but it allows for a clearer high-level view of how different external conditions evolve together across time. The dependent variable is the overall high school non-completion rate (*dropout_total_rate*). The explanatory variables were grouped according to the dimensions:

Economic factors: median income, low-income rate, and youth unemployment rate.

Social factors: population aged 15 to 19, attendance rate among youth aged 15 to 19, and the percentage of adults without a high school diploma.

Political factors: funding per student, a policy reform dummy, a spending shift dummy, and average secondary class size.

Table 1
Variables and Scope

Variable	Description	Role	Type	PESTEL category
dropout_total_rate	High school non-completion rate	Dependent variable	Numeric	Outcome
funding_per_student	Public funding per student	Explanatory	Numeric	Political
policy_reform_dummy	Reform period indicator	Explanatory	Binary	Political
spending_shift_dummy	Annual spending increase indicator	Explanatory	Binary	Political
secondary_avg_class_size	Average secondary class size	Explanatory	Numeric	Political
median_income	Median household income	Explanatory	Numeric	Economic
low_income_rate	Low-income prevalence	Explanatory	Numeric	Economic
youth_unemployment_rate	Youth unemployment rate	Explanatory	Numeric	Economic
demographic_total_15_19	Population aged 15–19	Explanatory	Numeric	Social
attendance_rate_15_19	Attendance rate for ages 15–19	Explanatory	Numeric	Social
pct_adults_no_hs_diploma	Adults without high school diploma	Explanatory	Numeric	Social

It is worth mentioning that political variables were further expanded during the project in order to capture structural conditions in a more complete way. This made it possible to distinguish between funding levels, formal policy changes, year-to-year spending movements, and class-size decisions, rather than relying only on financial indicators, as it had been initially proposed. Overall, the scope of the study remained intentionally exploratory. Given the short time series, the goal was to identify meaningful patterns and directional relationships, instead of establishing causality.

2.2 Data Cleaning

The cleaned dataset followed the general approach established in Part 1 of the project and was prepared through a series of practical steps to ensure consistency and usability across multiple public sources. All variables were first reviewed for data type consistency. Non-numeric entries in the original tables, including symbols used to represent missing or suppressed data, were converted into missing values where needed. Numeric fields were also explicitly formatted as numbers to ensure consistency and to prevent hidden data type issues during analysis.

Time information was then standardized into a single year field. This step was essential for merging data from different sources and making sure that all indicators aligned correctly over time. Only years with complete information across all selected variables were retained in the final analytical frame. On one hand, this reduced the dataset to only five yearly observations, but on the other, it ensured the analysis to be based on a consistent and comparable set of indicators.

Variable names were also simplified and standardized to improve readability and to make the workflow easier to manage in Python. This helped reduce confusion during data handling, visualization, and modeling.

Table 2

Data Preparation

Step	Action	Result
Merge datasets	Combined sources	Single dataframe
Missing values	Checked	None / handled
Type conversion	Numeric / int	OK
Feature engineering	Dummy reform	Added
Final dataset	Ready for analysis	Yes

Because the study includes only five yearly data points, no formal outlier detection or removal was performed. Instead, values were checked descriptively to confirm that they fell within reasonable ranges. In this context, preserving all available observations was more appropriate than applying aggressive cleaning rules. The final dataset was therefore restricted to Ontario and to the 2018-2022 period, in line with the original project proposal.

2.3 Analytical Methods

The analysis followed a step-by-step process, moving from simple inspection toward slightly more structured statistical techniques. Each stage was intended to build understanding gradually, rather than relying on a single method in isolation.

The starting point was descriptive analysis, including summary statistics, data structure checks, and basic inspection of value ranges. This helped confirm that the merged dataset was coherent and provided an initial sense of the scale and variation of each variable before interpretation actually began.

That was followed by trend analysis, carried out mainly through the use of line charts. Since the dataset covers only five years, visualizing yearly movement was more informative than focusing on distribution charts. These plots helped identify broad directions, timing, temporary disruptions, and shifts in the pattern of change across variables.

Next, basic statistical analysis was performed, including correlation matrices, scatter plots with fitted trend lines, and simple linear regressions. These resources were adopted to determine whether the patterns observed visually were also supported by linear relationships in the data. Given the limited number of observations, these findings were interpreted cautiously and treated primarily as directional signals rather than strong statistical evidence.

Comparative analysis was then used to make the results more concrete. Years were grouped into meaningful categories, such as pre- and post-policy reform periods and higher- versus lower-income years. These comparisons helped translate statistical relationships into more practical insights for the education context.

A final set of multiple regression models was estimated by dimension, with separate models for political, economic, and social variables, as well as a combined model. Again, it is important noticing that these models were not intended to support strong inference, but rather to explore how different groups of factors relate to non-completion rates when considered together.

Across the full process, the emphasis remained on clarity and interpretability. Again, because of the small sample size, the analytical strategy prioritized consistency across descriptive, visual, comparative, and regression-based findings. The results were then interpreted in relation to the project's broader PESTEL perspective, with particular attention to risks, opportunities, and strategic implications for the education sector.

3. Analysis and Results

This section brings together the descriptive, visual, and statistical analysis to understand how external PESTEL factors relate to high school non-completion over time. Given the small sample size, the analysis is intentionally cautious, focusing on patterns, alignments, and directional relationships rather than formal causal claims. Wherever possible, different methods (such as trends, correlations, and regressions) are used together, so that findings can be interpreted with a bit more confidence.

3.1 Initial Exploration

As previously mentioned, the final dataset contains five yearly data points for Ontario. Dropout rates range from 12.1% to 17.0%. Median income varies between 59,507 and 68,263 CAD, with funding per student ranging from 13,983 to 15,728 CAD. Social indicators such as attendance (around 60% - 63%) and the percentage of adults without a high school diploma (6% - 7%) remain relatively stable.

Even at this early stage, two distinct patterns become clear. Some variables (*income*, *funding*, and *adult educational attainment*) move gradually within a narrow range, suggesting more structural changes. Others, particularly *youth unemployment*, show sharper year-to-year variation, with a pronounced spike in 2020. This distinction becomes important later, as it helps explain why some relationships appear more stable while others are more reactive to short-term shocks.

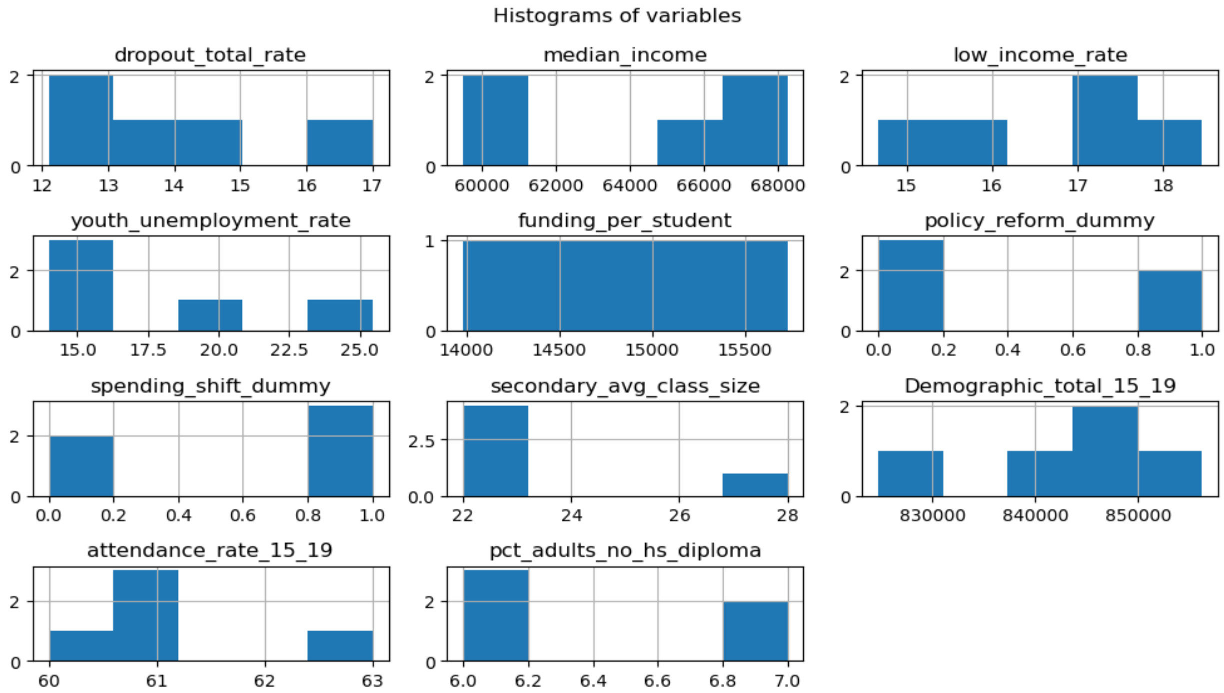
Table 3
Descriptive statistics

Variable	Mean	Std. Dev.	Min	Max	Interpretation
dropout_total_rate	14.22	1.90	12.10	17.00	Dropout rates vary moderately across years, with noticeable improvement in later years
median_income	64,422.80	3,869.55	59,507.00	68,263.00	Income shows steady growth over the period
low_income_rate	16.80	1.57	14.67	18.46	Poverty levels fluctuate but remain within a narrow range
youth_unemployment_rate	17.72	4.69	13.98	25.44	Youth unemployment shows strong variation, indicating economic shocks
funding_per_student	14,854.20	630.09	13,983.00	15,728.00	Public funding gradually increases over time
policy_reform_dummy	0.40	0.55	-	1.00	Reform period covers two of the five observations
spending_shift_dummy	0.60	0.55	-	1.00	Funding increased in most years
secondary_avg_class_size	23.80	2.39	22.00	28.00	Class size varies moderately across years
demographic_total_15_19	842,223.20	11,376.72	824,719.00	856,254.00	Youth population remains relatively stable
attendance_rate_15_19	61.20	1.10	60.00	63.00	Attendance rates show small variation
pct_adults_no_hs_diploma	6.40	0.55	6.00	7.00	Education level changes slowly over time

3.2 Descriptive Analysis

Aiming at understanding the basic structure and scale of the data, such as averages, spread and ranges, descriptive statistics were computed. With only five observations, each year carries significant weight to affect the overall pattern, so this step was mainly used to confirm that all variables fall within reasonable ranges and that there are no data quality issues.

Figure 1
Distribution of main variables used in the analysis



The results highlight differences in variability across variables. Economic indicators, especially *income* and *unemployment* show more movement across years, whereas demographic and structural variables remain relatively stable. Histograms were generated for completeness, but in such a small sample their interpretive value is limited. They were still useful to confirm scale consistency and verify that dummy variables were correctly coded as zeros and ones. Given these limitations, the analysis relies more heavily on time-based and relational visuals than on distribution-based ones.

3.3 Trend Analysis

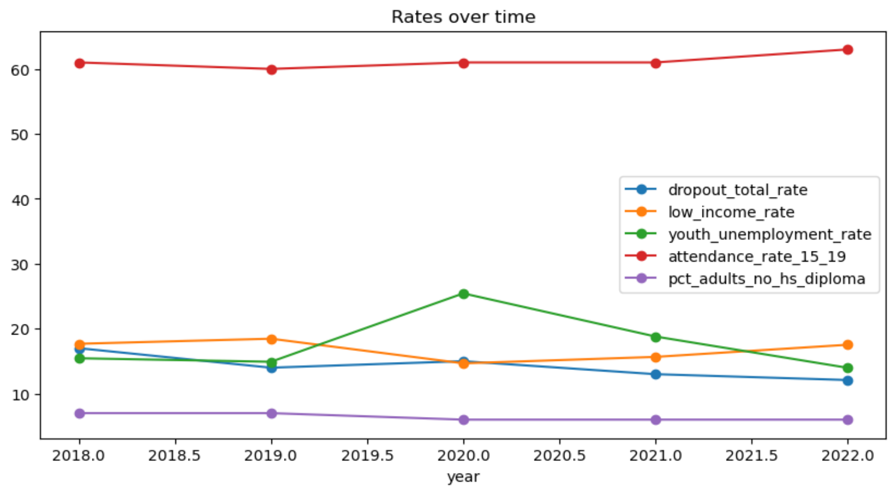
Because the dataset is a short annual time series, trend analysis plays a central role and the trend plots were one of the most informative parts of the workflow. They were used to observe how variables evolve over time and how these movements align with changes in dropout rates.

3.3.1 Evolution of Key Variables

The numeric variables were grouped by scale before plotting. This decision was important because variables such as *income*, *funding*, *attendance*, and *demographic counts* are measured on very different scales. Plotting them all in one graph would have hidden smaller series near the horizontal axis.

The rate-based line plots showed that the dropout rate generally declined across the five-year period, although there was a temporary increase in 2020. Attendance remained fairly stable, with a slight improvement by 2022. Youth unemployment displayed a very clear spike in 2020 before falling again, which suggests that short-term labour market stress may have affected educational outcomes during the middle of the period.

Figure 2
Trends of rate-based variables over time.

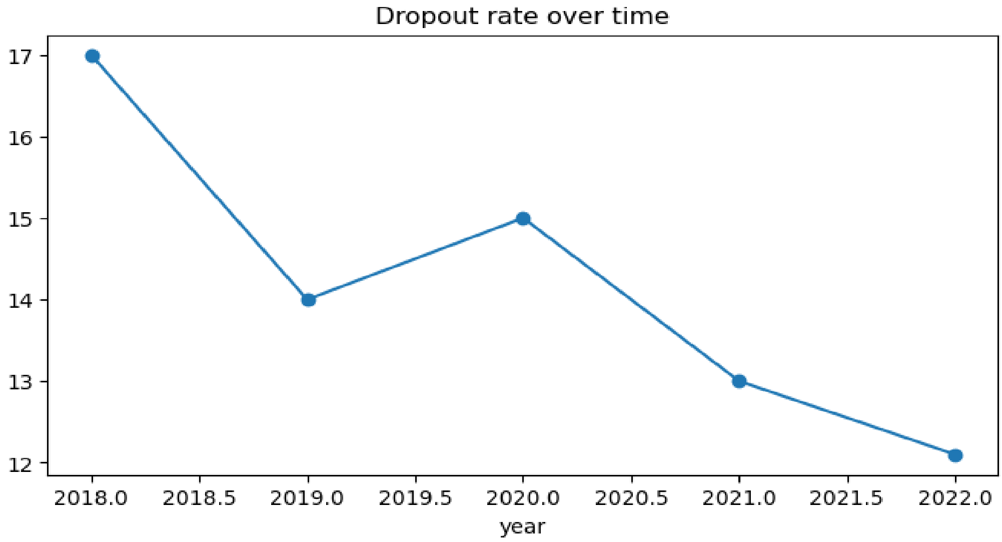


Note: *Variables are grouped by measurement scale to allow meaningful visual comparison.*

The *income* and *funding* plots showed a broadly positive direction over time. *Median income* rose steadily across the period, while *funding per student* increased overall after a dip in 2020. These movements matter because they suggest that the later years combined stronger household conditions with somewhat stronger institutional support.

The demographic and structural variables were more stable. The *population aged 15–19* did not show a dramatic directional shift, which suggests that cohort size alone is not enough to explain the observed change in dropout. Average class size showed some variation, but its pattern was weaker and less obviously aligned with dropout than the main economic and policy indicators.

Figure 3
Time trend of high school non-completion rate in Ontario.



Note: *Values represent yearly observations used in the regression analysis.*

3.3.2 Political and Policy Indicators

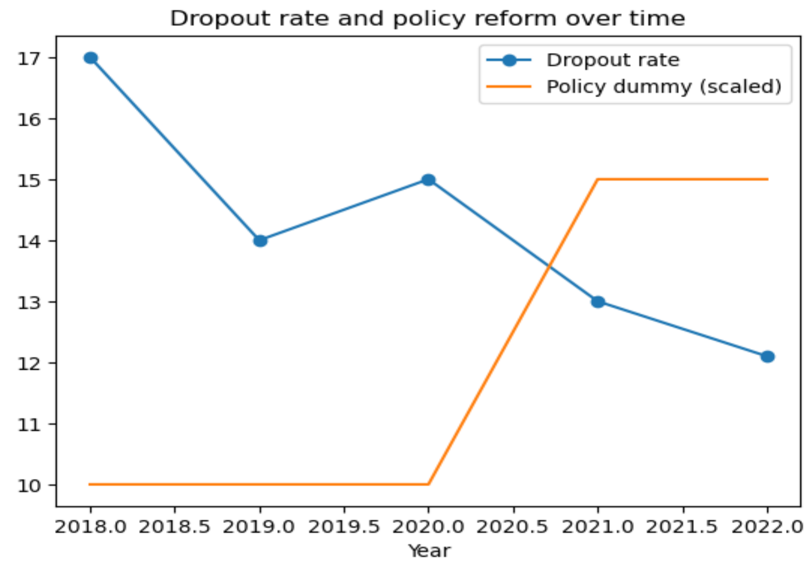
The two dummy variables were plotted separately because they represent structural shifts rather than continuous values. The *policy_reform_dummy* switches from 0 to 1 starting in 2021. This variable was designed to capture a major education reform period rather than the funding level itself. The *spending_shift_dummy* identifies whether *funding per student* increased relative to the prior year. Together, they allow the political dimension to reflect both structural reform and annual budget movement.

Plotting these policy variables helped clarify timing. It made the analysis more concrete by showing when the reform period begins and when spending increases occur relative to the observed movement in dropout.

3.3.3 Outcome Versus Policy

An additional line plot placed the dropout rate together with the reform indicator. However, this was not intended as proof of policy effect. Instead, the aim was to check visually whether the timing of the policy period coincides with a visible change in the outcome. The chart shows that the dropout rate is lower in the years after the reform indicator turns on, which is consistent with the expected direction of the political hypothesis (although the size of the dataset means that this must be treated as a descriptive pattern rather than a causal result).

Figure 4
Relationship between dropout rate *and* policy reform period



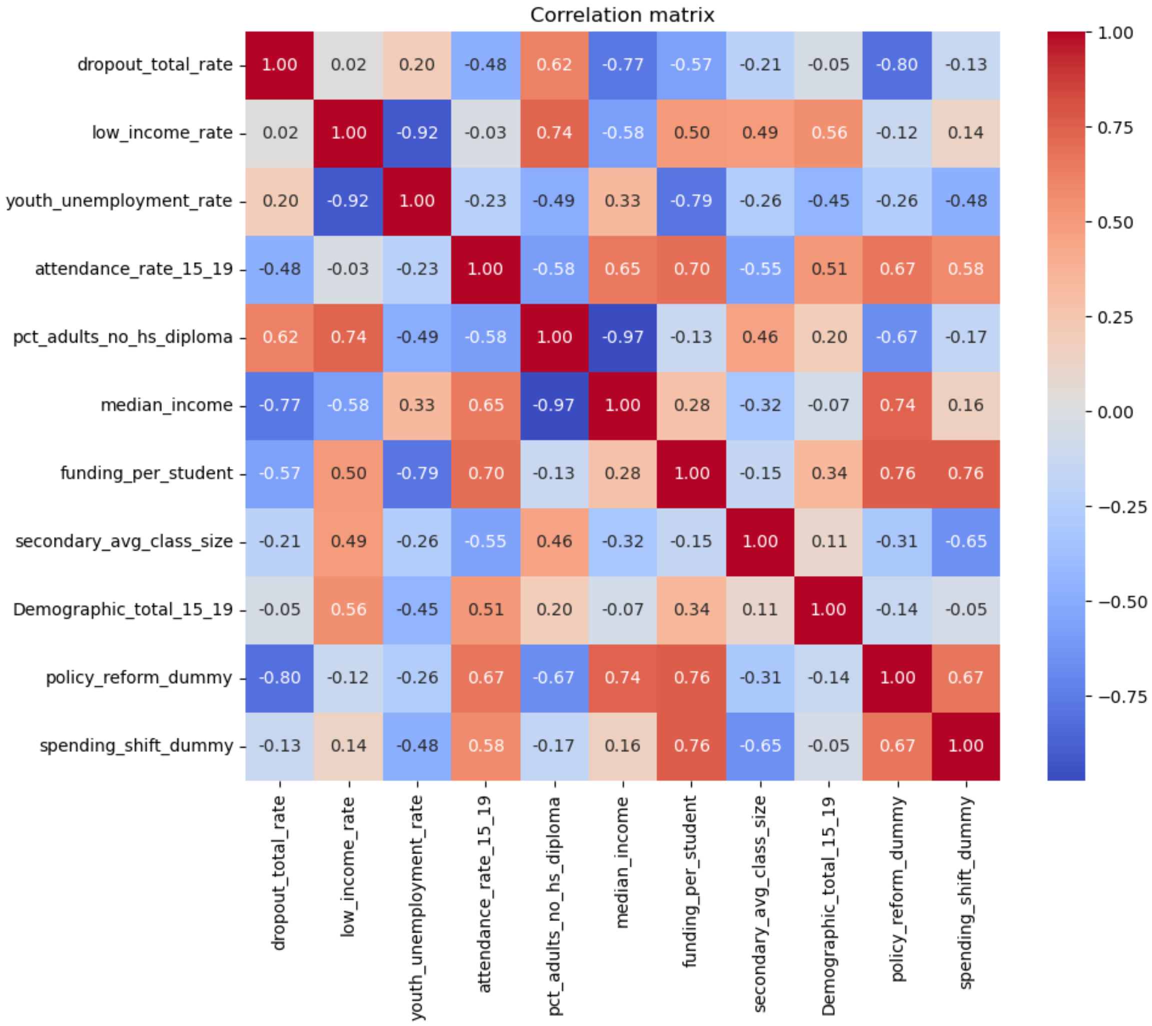
Note: *The policy reform dummy was scaled to allow visual comparison with the dropout rate.*

3.4 Correlation Analysis

A correlation matrix was computed to examine linear relationships among the main variables. Because the dataset contains only five observations, the correlations should be understood as directional signals rather than stable estimates.

The matrix suggests that *dropout* tends to be lower when *median income* and *funding* are higher, and higher when *socio-economic disadvantage indicators* are stronger. The reform dummy also showed a noticeable negative association with *dropout*. At the same time, the correlation matrix revealed strong relationships among some explanatory variables themselves, especially within the economic block.

Figure 5
Correlation matrix of variables used in the analysis

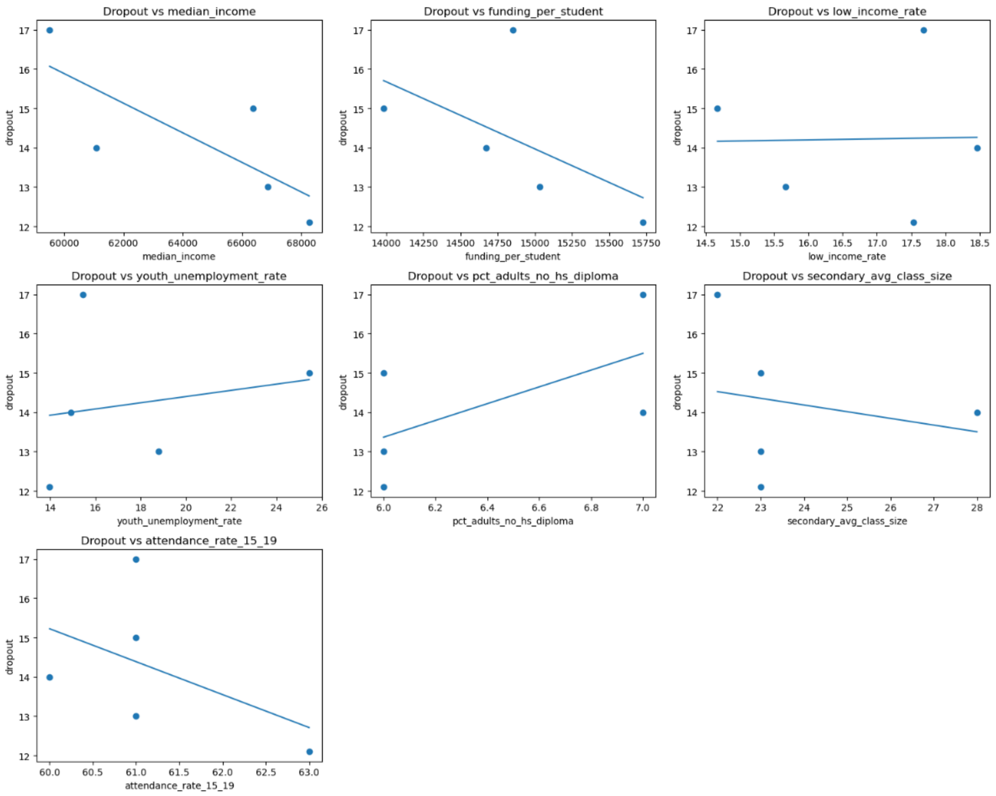


This matters for two reasons: first, it confirms that the sector appears sensitive to broader socio-economic conditions; second, it warns that multicollinearity could become a problem in later regressions if too many related variables are included at once.

3.5 Scatter Plot Analysis

Scatter plots with fitted trend lines were used to visually assess whether the relationships suggested by the correlation matrix are also visible in the raw data. These charts were especially useful because they showed that not all variables display the same strength of visual relationship with *dropout*. *Median income*, *funding per student*, and *attendance rate* displayed fairly clear negative patterns. In practical terms, years with stronger income, stronger funding, or stronger attendance tended to be associated with lower dropout.

Figure 6
Relationship between dropout rate *and selected predictors*



Note: *Lines represent simple linear fits used for visual inspection before regression modelling.*

By contrast, the percentage of adults without a high school diploma displayed a positive pattern, suggesting that a weaker adult educational background in the broader social environment coincided with higher dropout. Youth unemployment also showed a positive tendency, but that pattern was less smooth and clearly affected by the 2020 shock. Low-income rate and average class size appeared weaker and less consistent.

Therefore, the scatter plots played an important selection role, helping identify which variables would be most useful for the simple regressions and which variables were weaker candidates.

3.6 Regression Analysis

Moving beyond visual patterns and exploring the relationships more formally, regression analysis was used to examine how each PESTEL-related variable relates to changes in the dropout rate. Once more, it is worth mentioning that, given the very small sample size, these models are treated as exploratory rather than definitive, with the goal of identifying consistent directions and relative strength of associations rather than making strong causal claims. Both simple and multiple regression approaches are used, allowing for a clearer view of individual effects as well as how different factors interact when considered together.

3.6.1 Simple Regression Results

Simple linear regressions were estimated to evaluate the individual relationship between dropout rate and each explanatory variable. The single-variable results were broadly consistent with the earlier visual analysis. *Median income* had a negative coefficient and explained a meaningful share of variation on its own. *Funding per student* also showed a negative coefficient, although weaker than income. *Attendance rate* was negatively associated with dropout as expected. The percentage of *adults without a diploma* showed a positive relationship, which fits the idea that intergenerational educational disadvantage matters. *Youth unemployment* also had a positive coefficient, though its explanatory power was limited.

Table 4

Simple regression summary

Variable	Coefficient	Direction	R^2	Adj. R^2	p-value	Brief interpretation
median_income	-0.0004	Negative	0.590	0.454	0.129	Higher income is associated with lower dropout.
funding_per_student	-0.0017	Negative	0.322	0.096	0.318	Higher funding is associated with lower dropout, but the relationship is weaker.
youth_unemployment_rate	0.0790	Positive	0.038	-0.282	0.753	Higher youth unemployment is associated with higher dropout, but the fit is very low.
pct_adults_no_hs_diploma	21333	Positive	0.380	0.173	0.268	Lower adult educational attainment is associated with higher dropout.
attendance_rate_15_19	-0.8375	Negative	0.234	-0.021	0.409	Higher attendance is associated with lower dropout.
policy_reform_dummy	-27833	Negative	0.647	0.529	0.101	The post-reform period is associated with lower dropout levels.
spending_shift_dummy	-0.4667	Negative	0.018	-0.309	0.829	Year-to-year spending shifts alone show almost no relationship with dropout.

Note: Coefficient sign, model fit, and relationship direction are shown. Results are exploratory, not conclusive.

Among the policy variables, the *reform dummy* stood out. It showed a stronger relationship with dropout than the *spending-shift* dummy, which had almost no explanatory value on its own. This was an important finding because it suggested that the broader reform period may capture something more meaningful than a simple year-to-year spending increase.

3.6.2 Multiple Regression by PESTEL Dimension

In order to examine the joint relationship between dropout rates and the selected PESTEL indicators, multiple regression models were estimated by dimension. Separate models were built for the political, social, and economic components, followed by a combined model including one representative variable from each dimension. This structure enabled comparison of how each group of factors relates to the dropout rate while keeping the models more understandable.

A. Political Model:

The political regression included *funding per student* and the *policy reform dummy*. This model produced an R-squared of 0.652, which denotes a moderate fit relative to the other specifications. The coefficient for the reform dummy was negative, indicating lower dropout levels during the reform period, while the funding-per-student coefficient was very small and did not show a clear pattern in the model.

In practical terms, the political model suggests that the reform period may be more informative than annual funding changes alone. The general pattern points in a favourable direction, but the relationship is not as strong as in the economic model.

B. Economic Model

The economic regression included median income and youth unemployment rate. Among the three dimension-based models, this one performed best, with an R-squared of 0.817 and the lowest AIC of the separate models.

The coefficients followed the expected direction. Median income was negatively associated with dropout, indicating that higher income levels were linked to lower non-completion. Youth unemployment showed a positive coefficient, suggesting that labour market pressure may be associated with worse academic results. Compared with the political and social models, the economic block provided the clearest overall fit, which suggests that broader household and labour market conditions may play a particularly important role in explaining changes in dropout rates over time.

C. Social Model:

The social regression used the percentage of adults without a high school diploma and the attendance rate for ages 15–19. This model had the weakest performance among the three dimensions, with an R-squared of 0.404 and a negative adjusted R-squared.

Even so, the direction of the coefficients was still meaningful. The percentage of adults without a diploma was positively associated with dropout, while the attendance rate was negatively associated with dropout. These results are consistent with the wider descriptive analysis: a weaker educational background in the wider social environment tends to be associated with greater dropout, while stronger attendance is associated with lower dropout. Still, the overall explanatory power of the social model was limited.

D. Combined Model

A final combined model was estimated using one variable from each dimension: the policy reform dummy (political), percentage of adults without a high school diploma (social), and median income (economic). This model demonstrated the strongest overall fit, with an R-squared of 0.941, an adjusted R-squared of 0.764, and the lowest AIC among all specifications.

The combined results suggest that dropout rates are not related to only one type of external factor. Instead, the strongest fit appeared when political, social, and economic conditions were considered together. In this model, the reform dummy and median income remained negatively associated with dropout. The social variable changed direction compared with the social-only model, which suggests that its relationship becomes less stable once it is estimated together with variables from other dimensions.

E. Overall Comparison

Taken together, the regression results show a clear ranking across the models. The economic model was the strongest of the three separate dimensions, the political model showed moderate explanatory power, and the social model was weaker. The combined model outperformed all of them, which supports the idea that high school non-completion is determined by several external conditions acting at the same time rather than by a single factor in isolation.

Overall, these regression results are consistent with the descriptive patterns, correlations, and comparative tables observed earlier. They indicate that economic conditions alone provide the strongest signal, while a more complete understanding is obtained by jointly examining policies, the social context, and the economic context.

Table 5

Multiple regression model comparison

Model	Variables included	R^2	Adj. R^2	AIC	Interpretation
Political	funding_per_student, policy_reform_dummy	0.652	0.304	20.19	Moderate fit. Policy and funding show some association with dropout.
Social	pct_adults_no_hs_diploma, attendance_rate_15_19	0.404	-0.193	22.88	Weak fit. Social variables explain limited variation.
Economic	median_income, youth_unemployment_rate	0.817	0.633	16.99	Strongest single-dimension model. Economic conditions show high association with dropout.
Combined	policy_reform_dummy, pct_adults_no_hs_diploma, median_income	0.941	0.764	13.31	Best overall fit. Combining dimensions improves explanatory power.

Note. Models are compared by fit and explanatory power. Results are exploratory; the combined model fits best.

3.7 Comparative Analysis

This section compares outcomes across different contexts within the dataset, as a way to complement the regression results and provide a more concrete interpretation of the patterns observed. Rather than focusing only on statistical relationships, these comparisons allow to observe how dropout rates and related indicators behave under different policy and economic conditions. By grouping the data into meaningful segments, such as *before* and *after the policy reform*, and *higher-* versus *lower-income years,* it becomes easier to see how combinations of PESTEL factors align with changes in student outcomes, and whether the patterns identified earlier hold when viewed from a more practical perspective.

3.7.1 Comparison *Before* and *After* Policy Reform

The data were split into *pre-reform* and *post-reform* periods using the *policy_reform_dummy* in order to better understand the political dimension. This comparison helps move from statistical associations to a more concrete interpretation of the policy environment.

The post-reform years show a lower average dropout rate than the pre-reform years. They also show higher funding per student and slightly higher attendance rates. Median income is also higher in the later period, while the percentage of adults without a high school diploma is slightly lower.

Table 6

Comparison before vs after policy reform

Variable	Before reform	After reform
Dropout rate	15.33	12.55
Median income	62324.67	67570.00
Low-income rate	16.94	16.60
Youth unemployment	18.60	16.39
Adults without HS diploma	6.67	6.00
Funding per student	14503	15381
Attendance rate	60.67	62.00

Note. Values are averages before and after reform. The comparison is descriptive and does not imply causality.

These results are consistent with the regression models, where the reform dummy showed a negative coefficient and the political model displayed moderate explanatory power. The comparison suggests that the later years were more favourable for student outcomes, but the improvement does not appear to come from policy alone.

Economic conditions also improved during the same period, which means the reform coincided with a broader change in the external environment. For this reason, the most careful interpretation is that the reform period aligns with better outcomes, but the effect likely reflects a combination of political and economic factors rather than a single cause.

3.7.2 Comparison *High-* Versus *Low-Income Years*

A second comparison was created by splitting the years into higher-income and lower-income groups based on the sample mean of median income. This allows examining how broader economic conditions relate to changes in dropout rates.

The comparison shows that higher-income years tend to have lower dropout rates, lower poverty levels, and slightly higher attendance. Funding per student is also somewhat higher during these years. These patterns are consistent with the regression results, where the economic model showed the strongest overall fit among the three dimensions.

Table 7

Comparison by income level

Variable	Low-Income Years	High-Income Years
Dropout rate	15.50	13.37
Low-income rate	18.07	15.95
Youth unemployment	15.18	19.41
Funding per student	14763	14915
Attendance rate	60.50	61.67

Note. Years are grouped by income level. Higher-income years show lower dropout and slightly higher attendance.

One result in the above table is less straightforward. Youth unemployment is slightly higher in the high-income group. Given the very small number of observations, this likely reflects short-term variation in the labour market rather than a stable relationship. The regression model also showed that unemployment does not move perfectly with income over this short period.

Overall, the comparison supports the idea that economic conditions are closely related to dropout trends. When household income is stronger and poverty is lower, the education system also tends to show better outcomes. This result is consistent with the regression analysis, where the economic variables provided the highest explanatory power.

4. Environmental Impact Analysis

This section builds on the previous analysis by translating the statistical findings into practical insights for decision-making. Rather than introducing new results, it focuses on interpreting how external conditions shape educational outcomes in Ontario. Using the PESTEL perspective, the goal is to move beyond identifying patterns in the data and instead understand what these patterns mean in terms of risks, opportunities, and the broader environment in which the sector operates.

In order to bring the results together in a structured way, the main variables were organized into a PESTEL summary table, more specifically, into the Political, Economic, and Social dimensions. The goal of the table was not to add new statistics, but to translate the evidence into sector-level risks and opportunities.

Table 8

Enviroment Impact Analysis

Dimension	Variable	Main pattern observed	Risk / Opportunity for the sector
Political	policy_reform_dummy	Dropout lower after reform	Opportunity if policy stability continues
	funding_per_student	Higher funding linked to lower dropout	Opportunity through public investment
Economic	median_income	Higher income linked to lower dropout	Risk if economic conditions weaken
	low_income_rate	Higher poverty linked to higher dropout	Risk for vulnerable communities
	youth_unemployment_rate	Higher unemployment linked to higher dropout	Risk during labour market shocks
Social	pct_adults_no_hs_diploma	Lower adult education linked to higher dropout	Long-term social risk
	attendance_rate_15_19	Higher attendance linked to lower dropout	Opportunity through student engagement

4.1 Political Dimension

The political environment is primarily reflected through the policy reform indicator and funding per student, both of which show a consistent relationship with dropout rates. The post-reform period is associated with lower average dropout levels compared to earlier years, and higher funding tends to align with better outcomes. While this does not establish direct causality, the pattern suggests that a stable and supportive policy framework creates more favourable conditions for student retention.

From a decision-making perspective, this points to a clear opportunity. Maintaining consistent policy direction and sustained public investment appears to strengthen the system's ability to support students over time. The reform period aligns with improved outcomes, indicating that structural policy changes can contribute to a more supportive educational environment.

At the same time, this also highlights a risk. If funding becomes unstable or policy direction shifts frequently, the progress observed in recent years may not hold. The sector appears sensitive to its political context, meaning that instability at this level can translate into weaker outcomes at the student level.

4.2 Economic Dimension

The economic variables (*median income*, *low-income rate*, and *youth unemployment*) show the strongest and most consistent relationships with dropout rates across the analysis. Higher income levels are associated with lower dropout, while higher poverty and unemployment tend to align with worse outcomes. This pattern appears consistently across trends, correlations, and regression models, reinforcing the strength of the economic signal.

This suggests that dropout is not only a function of what happens within schools, but also of the financial stability of households. When families experience less economic stress, students are more likely to remain engaged in education. Conversely, periods of economic disruption can place upward pressure on dropout rates, revealing how vulnerable the system is to broader economic shocks.

From an environmental perspective, this represents a key risk. Even well-functioning schools may struggle to offset the effects of worsening economic conditions. At the same time, stronger economic environments may create a significant opportunity, as improvements in income and reductions in poverty appear to support better educational outcomes.

4.3 Social Dimension

The social dimension captures the broader context in which students live and make decisions, including *attendance*, *adult educational attainment*, and *youth unemployment*.

Attendance emerges as one of the most consistent and practically relevant variables. Higher attendance is clearly associated with lower dropout, making it both an early signal of disengagement and a direct lever for intervention. This creates a concrete opportunity for the sector, as attendance can be monitored closely and addressed through targeted programs aimed at improving student engagement.

At the same time, the percentage of adults without a high school diploma points to a deeper, long-term structural influence. Communities with lower levels of educational attainment tend to face greater challenges in supporting student completion, suggesting the presence of intergenerational effects. This represents a more persistent risk, one that cannot be addressed through short-term interventions alone.

Youth unemployment, in turn, further reinforces the importance of social and labour market stability. Fluctuations in employment conditions can influence student decisions, highlighting how closely educational outcomes are tied to the broader social environment.

4.4 Overall Environmental Interpretation

Considered together, these findings show that dropout rates are shaped by the interaction of political, economic, and social factors rather than by one single variable. The sector does not operate in isolation; it is embedded in a wider system where external conditions continuously influence outcomes.

For decision-makers, this means that risks and opportunities are interconnected. Economic stress, policy instability, and adverse social conditions can reinforce one another, increasing the likelihood of negative outcomes. At the same time, improvements across these dimensions can work together to create a more supportive environment for student success.

This also highlights an important opportunity. While some of these factors lie outside the direct control of the education system, they are not beyond influence. Strategic coordination with other areas, such as economic development, social services, and community programs, can help address the root causes of dropout rather than only its symptoms within schools.

Overall, the analysis suggests that effective strategies must extend beyond the classroom. Understanding and responding to the broader environment is essential for sustaining progress and improving long-term educational outcomes.

5. Strategic Insights

This section focus on what decision-makers can do in response to the patterns identified. Rather than revisiting the interpretation of external factors, it seeks to translate those insights into concrete strategic considerations, highlighting where action can mitigate risk, strengthen outcomes, and take advantage of favourable conditions. The emphasis here is not only on understanding the environment, but on responding to it in a coordinated and practical way.

The findings from this report are important for they push the conversation beyond simple school-level fixes. There are a few key takeaways that should shape how those making decisions in education policy or administration approach the problem of high school non-completion.

First, decision-makers must understand that education policy is most effective when it is consistent and adequately funded. The reform period in our data aligns with better outcomes, and funding per student moves in the same direction. This suggests that chopping and changing direction, or starving the system of resources during tough times, is likely to make a difficult problem worse. A stable, long-term policy environment creates the conditions for success.

Second, and perhaps most importantly, the evidence shows that you should not separate the education sector from the economic reality of families. The household economic conditions are so strongly connected to dropout rates that it would be a mistake to treat this as purely a "school problem". When the economy is strong and household incomes are higher, the sector performs better. When there is an economic shock or high poverty, the sector struggles. For a decision-maker, this means that strategies to reduce dropout must include efforts to support economic stability in communities. It also means that education departments need a seat at the table when broader economic policies are being discussed.

Third, the analysis points to attendance as a critical and actionable signal. While it's influenced by deeper forces, it's also something schools can track and respond to in real-time. A focus on improving attendance is not just about discipline, but a direct strategy for reducing dropout. It is perhaps one of the most effective short-run levers available to school leaders.

Finally, the interconnected nature of these findings means that risks and opportunities are two sides of the same coin. A negative environment (with a weak economy, funding cuts, and high poverty) creates a powerful drag on the system, where risks compound each other. Conversely, a positive environment (with stable policy, rising incomes, and strong engagement) creates a virtuous cycle. Therefore, the strategic insight here is to avoid “siloed” thinking. The most effective solutions will probably be those that look at the whole picture, addressing the political, economic, and social factors together, rather than trying to fix the dropout problem one student at a time.

6. Limitations

The findings, while revealing, must be interpreted considering several critical limitations, which are not isolated issues, but rather overlapping constraints that shape how results should be interpreted.

The most obvious one is the sample size. With only five annual observations, statistical power is extremely limited, and every estimate becomes highly sensitive to individual years, especially atypical ones such as 2020, which coincides with the COVID-19 pandemic, a significant external shock that affected both the education system and broader socioeconomic conditions. School closures, shifts to remote learning, and disruptions to household stability may have influenced student engagement and completion patterns in ways that are not fully captured by the variables included in the dataset. At the same time, several of the macro-level indicators used in the analysis were also affected, potentially introducing short-term distortions into the observed relationships.

In such a small dataset, a single data point can disproportionately influence correlations and regression outputs, which makes the results inherently unstable. This is why any quantitative findings are better understood as exploratory cues rather than confirmatory evidence, since there is not enough information to clearly distinguish signal from noise. For example, even if income genuinely affects dropout rates, the model might not “see” that effect, as it might look non-significant just because the dataset is too small.

At the same time, the data are aggregated at a level that may mask a significant degree of its underlying complexity. Provincial averages for Ontario inevitably conceal regional disparities, differences across school boards, and the realities of individual schools and students. What appears as a single trend at the provincial level may, in fact, be driven only by some specific areas, while others follow entirely different patterns. This may incur in aggregation bias and, more importantly, in the ecological fallacy, meaning that relationships observed in the aggregate cannot be assumed to hold at the individual level. The analysis observes patterns in “years”, not in “students”.

There is also an inherent constraint in the relationship between the variables, as many of the PESTEL factors considered – especially economic and social indicators – evolve together over time, which may introduce multicollinearity. In practical terms, this makes it hard to isolate the independent effect of each variable with a high level of confidence. The pre- and post-reform comparison is an example, where multiple changes occur simultaneously and make attribution ambiguous. Because the variables are not separable, the results become difficult to interpret as the coefficients may reflect joint trends rather than individual causal effects. In fact, the issue is taken even further in the multiple regression model presented, which is likely overfitted: with no residual degrees of freedom, it cannot support meaningful inferential interpretation.

Another layer of complexity emerges when considering the direction of relationships. The analysis implicitly assumes that political and economic conditions influence dropout rates, yet this relationship may not be strictly one-way. Endogeneity must be a real concern here, since persistent dropout patterns could themselves shape policy decisions or economic investment over time, creating feedback loops that the current dataset cannot clarify (this reinforces the need for caution when interpreting any apparent associations as causal).

Moreover, there are important limitations in terms of measurement and scope. The analysis framework does not take into account all PESTEL components. The technological, environmental, and legal dimensions are absent, so this project does not fully operationalize the context, and although this does not invalidate the findings, it calls for a more precise framing of what is being measured (only political, social and economic factors).

At last, it is mandatory mentioning the issue of data quality itself. One of the dropout observations, for 2022, is not an actual recorded value but a statistical estimate derived from a forecasting approach. This may introduce measurement error into the dependent variable, and because that is a variable placed at the center of the analysis, its uncertainty propagates through every correlation and model built upon it. Combined with the small sample size, this only amplifies the fragility of the findings.

7. Future Improvements

The most urgent improvement would be to extend the time horizon of the dataset. With only a small number of annual observations, the analysis remains primarily descriptive, capturing how variables move together rather than allowing for more robust statistical inference. Expanding the dataset to cover a longer period (ideally a range of 10 to 15 years) would allow for more robust and appropriate time-series analysis, making it possible to better isolate the effects of policy changes (such as the 2021 reform) while accounting for underlying trends. A longer series would also reduce the sensitivity of the results to isolated disruptions, allowing patterns to emerge more consistently over time.

At the same time, increasing the granularity of the data would significantly strengthen the analysis. Moving beyond provincial aggregates to observations at the level of school boards, census divisions, or even individual schools and students would not only increase the sample size, but also make it possible to test whether the patterns observed at the provincial level actually hold within individual units. This would naturally support the use of panel data methods, which can control for unobserved, time-invariant differences across regions and improve the scenario for causal claims to be eventually made. Greater granularity would also enhance interpretability, bringing the analysis closer to the actual realities behind the data.

There is also clear room to broaden the PESTEL framework itself. As it stands now, the model captures political, economic, and social dimensions only, but leaves technological, environmental, and legal factors largely unobserved. Incorporating indicators such as digital access, device availability, or infrastructure readiness would allow the analysis to account for technological disparities that may shape educational outcomes. Environmental variables, for instance those tied to place-based disruptions, could add further explanatory power, while legal indicators related to compulsory schooling policies, disciplinary frameworks, or student rights would enrich the picture. In addition, the political variables could move beyond simple binary indicators toward more nuanced representations, such as timing of implementation or qualitative coding of policy design.

As the data structure becomes richer, the choice of analytical methods should evolve accordingly. While the current approach provides a useful baseline, more sophisticated techniques (such as multivariate models, panel regressions, or interrupted time series) would be better suited to capturing the complexity of a longer and more detailed dataset. With sufficient observations, it would also become feasible to explore non-linear relationships. For instance, the analysis could investigate whether there is a tipping point in per-student funding below which dropout rates begin to increase disproportionately, revealing dynamics that linear models might overlook.

Finally, complementing the quantitative analysis with qualitative insights would add an important layer of depth. Patterns observed in the data could be contextualized through policy analysis, administrative records, or interviews with school administrators, teachers, and students, especially in regions of higher risk profiles. This approach would not only help explain why certain years or reforms align with stronger or weaker outcomes, but would also relate the statistical findings more closely with the institutional and human realities they reflect.

8. Conclusion

This project set out to examine how broader external conditions relate to high school non-completion in Ontario, with the aim of supporting more informed and context-aware decision-making. Using a PESTEL-informed perspective and a stepwise analytical approach, the findings consistently point to a central conclusion: dropout is not an isolated educational issue, but a phenomenon shaped by the interaction of political, economic, and social forces that define the environment in which students and schools operate.

Among these dimensions, economic conditions emerge as apparently the most influential and consistent factor. The strong alignment between household financial stability and student outcomes (particularly through indicators such as income, poverty, and youth unemployment) suggests that non-completion cannot be fully understood, or effectively addressed, without considering the broader socio-economic context. This shifts the framing of the issue beyond schools alone, highlighting the foundational role of economic stability in supporting student retention.

Political factors also seem to play a meaningful role, particularly through policy stability and sustained investment. The observed improvements associated with the post-reform period and higher per-student funding suggest that consistent and well-supported policy environments can create the conditions necessary for progress. Social variables, while more limited in explanatory power, provide important complementary insight. In particular, attendance stands out as a practical and actionable lever, functioning both as an early warning signal and a direct point for intervention, while broader indicators such as adult educational attainment reflect longer-term structural influences.

When considered together, these findings reinforce the idea that risks and opportunities in the education sector are deeply interconnected. Economic hardship, policy instability, and adverse social conditions can combine to amplify dropout risk, while improvements across these areas can generate cumulative positive effects. As a result, effective responses require moving beyond fragmented or school-centric approaches. As the literature widely shows, education policy is most impactful when coordinated with broader economic and social strategies, recognizing that student outcomes are shaped by a wider system.

At the same time, these conclusions must be interpreted with appropriate caution. As previously mentioned, the analysis is constrained by a small sample size, the aggregated nature of provincial data, and the absence of some PESTEL dimensions. These limitations mean the findings should be understood as exploratory and directional, offering insight into how different factors relate rather than providing definitive causal claims. Future research would benefit from expanding the time horizon, incorporating more granular data, and broadening the analytical scope to build a more robust understanding.

Despite these constraints, the analysis still expects to provide a valuable perspective, by underscoring the importance of viewing education as part of a broader system and highlighting the need for integrated, long-term strategies. After all, addressing high school non-completion seem to demand not only improving what happens within schools, but also engaging with the wider economic, political, and social environment that shapes student outcomes over time.

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